08-17 第1篇

阅读理解（说明文） How Do Birds Find Their Way When They Fly South?

① Every autumn, if you look up at the sky just before sunset, you may see long lines of birds flying in a neat "V" shape toward the south. Some of these birds, like the little Arctic **tern**, will travel more than **70,000 kilometres** in a year, from the cold far north to the far south and back. They do this without any map, any phone signal, or any teacher. How can such small creatures find the right road across half of the Earth? Scientists have been studying this question for almost a hundred years, and the answer is made up of several amazing tools.

② The first tool is the **sun**. Many **migratory birds** leave during the day, and they use the sun like a large, slowly-moving compass. They know roughly where the sun should be at each hour of the morning or afternoon. When the sun is where they expect, they know they are flying the right way. Their tiny internal clocks keep track of time, so the moving sun does not confuse them.

③ The second tool is the **stars**. Some birds fly mainly at night, when the weather is cooler and there is less wind. Scientists once placed young night-flying birds inside a planetarium — a big room where false stars are shown on the ceiling. The birds always turned toward the true south even though they were inside a closed room. This means they are born with some basic "star maps" in their brains; they do not need a teacher to show them.

④ The third tool is perhaps the most surprising: the **Earth itself acts like a giant magnet**. Birds seem to have very small **magnetic particles** in their eyes and in their beaks. These particles help them feel the Earth's magnetic field, so they always know which way is north and which way is south, even on cloudy nights when no star or sun can be seen. In this way, even a small thunderstorm in the afternoon cannot make them completely lost.

⑤ The fourth tool is **landmarks**. On long journeys, rivers, mountains, coastlines, and lakes act like road signs. A young bird flying south for the first time often follows an older one. The young bird remembers the shape of the big river, the curve of the coast, the snow-covered mountain in the middle. The next year, when the young bird travels alone, these memories guide it like a silent map.

⑥ Finally, birds have **very good eyes** and a strong sense of the weather. They can feel the wind changing hours before a storm, and they often wait on the ground until the

wind is right. Some species even fly higher on clear days and lower over the sea, using the air like a skilled pilot.

⑦ Of course, human cities today are making their journey harder. Bright lights from our tall buildings can confuse night-flying birds; wide glass windows are sometimes invisible to them. In more than twenty cities around the world, people have begun to **turn off unnecessary lights for two weeks every spring and autumn** to help the birds find their way safely.

⑧ Next time you hear the quiet cries of wild geese in the October evening, remember: above our rooftops, a small living compass is working with the sun, the stars, the Earth's magnetism, and memory — all at the same time — to guide these birds across the world.

注释： tern 燕鸥；migratory 迁徙的；planetarium 天文馆；magnetic 磁的；landmark 地标

根据文章内容，选择最佳答案。

1. Which bird is mentioned in the passage as travelling more than 70,000 kilometres in a year?

A. The sparrow.
B. The Arctic tern.
C. The parrot.
D. The pigeon.

2. How do day-flying migratory birds mainly find their way according to the passage?

A. They wait for a human guide.
B. They use the sun like a compass, with the help of their internal clocks.
C. They listen to the radio in the sky.
D. They follow a leader bird with a GPS phone.

3. What did the planetarium experiment show?

A. Young night-flying birds are born with a basic understanding of stars in their brains.
B. Birds need a teacher to learn how to use the stars.
C. Star maps are useless to birds.
D. Birds cannot see stars at all.

4. Why are the Earth's magnetic particles in birds' eyes and beaks useful?

A. They help the birds see in strong sunlight.
B. They help birds sense the Earth's magnetic field, so they still know north and south on cloudy nights.

C. They make the birds' feathers softer and warmer.
D. They make the birds lighter in the air.

5. What does the passage most likely want us to do?

A. Stop all birds from flying south.
B. Pay more attention to how our city lights and glass affect migratory birds, and be willing to help them.
C. Build higher and brighter buildings in every city.
D. Keep all migratory birds at home as pets.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第①段 "the little Arctic tern, will travel more than 70,000 kilometres"
【解析】原文明确提到北极燕鸥年飞 7 万多千米。A、C、D 均未在原文提及此数据。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第②段 "they use the sun like a ... compass ... Their tiny internal clocks keep track of time"
【解析】原文明确白天靠太阳+生物钟。A、C、D 均无文中依据。故选 B。

3.
【答案】A
【题型】推理判断题
【定位】第③段 "they are born with some basic 'star maps' in their brains"
【解析】实验结论表明夜飞幼鸟天生具备基本的"星图"。B 与原文相反；C、D 亦与原文矛盾。故选 A。

4.
【答案】B
【题型】细节理解题
【定位】第④段 "These particles help them feel the Earth's magnetic field, so they always know which way is north and which way is south, even on cloudy nights"
【解析】磁粒子帮助感知地磁场，阴天也能辨向。A、C、D 均无文中依据。故选 B。

5.
【答案】B
【题型】写作意图题
【定位】第⑦段 & 第⑧段

【解析】作者提到 20 多座城市关灯帮鸟，并在结尾让读者欣赏鸟类的"活罗盘"，表达应关心并帮助迁徙鸟。A、C、D 均与作者立场相反。故选 B。